

Signal Audit

Dataset and loading

The dataset contains images of two pistachio classes: **Kirmizi** and **Siirt**. Before loading the images, the code checks that both dataset directories exist. Both directories were successfully found.

Since the complete dataset contains more than 2000 images, this audit uses a small subset of five images from each class. This gives a total of 10 images and makes the signal analysis faster while still allowing the structure and properties of the image data to be inspected. Each image is loaded with PIL, converted to RGB, and stored as a NumPy array.

Format, array shape, data type, units, range and byte size

The file format was checked using the first bytes of each file rather than relying only on the filename extension. All 10 tested images were identified as JPEG files.

Each image has an array shape of (600, 600, 3). The first two dimensions represent the image height and width, while the final dimension contains the three RGB colour channels. The data type is uint8, meaning that each colour-channel value is represented by an unsigned 8-bit integer.

Pixel values therefore have a possible range from 0 to 255. The loaded subset was verified to contain this complete range in all three RGB channels. These values represent digital intensity levels rather than a physical measurement unit.

A single uncompressed RGB image requires:

$$600 \times 600 \times 3 = 1,080,000 \text{ bytes}$$

in memory. For comparison, the first JPEG image occupies only 22,307 bytes on disk. This difference occurs because JPEG uses compression, while the NumPy representation stores every RGB sample directly.

Sampling and spatial spacing

An image can be considered a two-dimensional sampled signal. In this dataset, one spatial sample corresponds to one pixel, producing a sampling grid of 600×600 pixels. If the RGB channels are flattened into a one-dimensional array, one image contains 1,080,000 channel values.

The JPEG metadata reports a resolution of 96 DPI, corresponding to approximately 3.78 pixels/mm . However, this should not be interpreted as a reliable physical sampling rate. DPI metadata can describe how an image should be displayed or printed and does not prove that the photographed object was sampled at this physical spacing.

Because the dataset provides no camera calibration, field of view, or known physical dimensions, the physical size represented by one pixel cannot be determined. Therefore, frequency measurements in this audit are expressed in **cycles per pixel** rather than cycles per millimetre.

Spatial and frequency-domain views

The image was first converted to grayscale by averaging its RGB channels. The spatial-domain representation shows intensity as a function of the image's x and y pixel coordinates.

A two-dimensional Fast Fourier Transform (FFT) was then applied to produce a frequency-domain representation. The centre of the shifted FFT corresponds to low spatial frequencies, which represent slowly changing or large-scale image structures. Frequencies farther from the centre represent faster spatial changes, such as edges, texture, and fine detail. The frequency axes are expressed in cycles per pixel because no reliable physical spatial calibration is available.

Gaussian filtering experiment

Before applying the filter, the prediction was that Gaussian smoothing would reduce high spatial frequencies while preserving the general shape and large-scale colour structure of the pistachio.

A Gaussian filter with $\sigma = 5$ was applied independently to the three colour channels. The resulting image is visibly smoother, while the overall object shape remains recognizable. The difference image shows that most of the removed information occurs around edges and fine details.

The FFT of the filtered image confirms the prediction. After Gaussian filtering, the spectrum is more concentrated around the centre, while frequencies farther from the centre have lower magnitude. This occurs because a Gaussian blur behaves as a low-pass filter: it preserves low-frequency components while attenuating high-frequency components. While this is harder to see because of the black background being black, the difference photo showcases this especially as the high frequencies are blurred.

Resampling experiment

A resampling experiment was performed by reducing an image from 600×600 pixels to 75×75 pixels, corresponding to a downsampling factor of eight. The image was then enlarged back to its original dimensions to make the information loss visible.

The experiment produced a maximum pixel error of 190.7 out of 255 and a mean error of 2.51. Fine details were lost, and edges became more blurred or block-like. This demonstrates an important limitation of spatial resampling: once the number of samples is reduced, fine spatial information cannot be fully reconstructed. High-frequency details that were present in the original 600×600 image are no longer represented by the lower-resolution signal.

Baseline representation for P2

For P2, the baseline representation is the decoded, uncompressed RGB uint8 array. For the 10-image subset, the baseline has shape (10, 600, 600, 3) and occupies 10,800,000 bytes in memory, or 1,080,000 bytes per image.

This representation acts as the reference against which later encoding or compression methods can be compared. Storage size and reconstruction differences can therefore be measured relative to the same decoded RGB representation.

Visual informatics and implementation concerns

One visual-informatics limitation is the absence of physical calibration. Although the images contain DPI metadata, there is insufficient information to determine the actual physical distance represented by one pixel. Consequently, the analysis can reliably describe frequencies in cycles per pixel, but not in physical units such as cycles per millimetre.

There are also implementation and data-integrity concerns. The loader initially selects files using their extensions, so malformed or incorrectly labelled files could be encountered. Checking file signatures and allowing PIL to decode the files provides additional validation. Furthermore, the dataset images appear to have undergone preprocessing, but the exact preprocessing pipeline is not documented. Operations such as resizing, cropping, colour adjustment, or compression may therefore already have altered the signals before this analysis was performed.

Work distribution

The group worked in a pair, not mainly one doing only one task, but all tasks were solved together, then later Vinjar cleaned up and added some extra content to the notebook. The Document was mainly written by Ole, with extra input by Vinjar.

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